

G.H.D.B. Ashen

Undergraduate Software Engineer

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PROFESSIONAL SUMMARY

Innovative Software Engineering Undergraduate specializing in modern web architectures and serverless portability across Azure, AWS, and GCP. Experienced with using React, Next.js, and Express.js with a deep understanding of RESTful API design and Postman testing. Demonstrated ability to modernize complex workflows through Docker containerization and efficient database management in MySQL and MongoDB. Highly collaborative leader with success in national-level technical competitions.

PROJECT EXPERIENCE

Banquet Catering Management System (BCMS)-Group Project, React.js, Express.js, MySQL (February 2025 - May 2025)

- Engineered a full-stack web platform to streamline banquet and catering operations
- Implemented modules for booking management, employee scheduling, and customer interaction tracking
- Designed a responsive frontend using React and structured relational data handling with MySQL

FOT Canteen management System in Hacktrail 2.0-Group Project,

React.js, Express.js, MySQL (August 2025 - September 2025)

- Developed a web application to modernize canteen operations and ordering workflows
- Optimized the food ordering pipeline to improve user experience and efficiency
- Managed end-to-end feature lifecycle from menu creation to order processing

Health Tracker (FitTrack)-Group Project, Java Spring Boot, React (October 2025)

- Architected a scalable health monitoring system with a focus on backend performance
- Implemented RESTful backend services using Spring Boot to manage user health data
- Integrated frontend and backend components for real-time data interaction

Retrieval Augmented AI Advisor, Next.js, Python, FastAPI, Docker, Jenkins, CI/CD (February 2026 - March 2026)

- Developed a Full-Stack RAG Platform: Built a "RAG-as-a-Service" application using FastAPI and Next.js that allows users to create custom, data-driven chatbots.
- Engineered AI Retrieval Pipeline: Implemented a pipeline in Python using Groq's Llama-3 models and vector database searches to provide context-aware responses with automated source citations.
- Automated DevOps Workflow: Containerized the application with Docker and established a CI/CD pipeline using Jenkins to streamline production deployments.

Roadside Assist Pro, React, Express, MongoDB, OpenStreetMap (February 2026)

- Built a live coordination system using Socket.io for instant messaging, featuring persistent chat history and real-time event notifications for users and mechanics.
- Designed a role-based platform with React and TypeScript, implementing secure routing and separate, interactive dashboards for complex state management between mechanics and customers.
- Leveraged map-based discovery to enable location-based matching, allowing for efficient customer-to-mechanic coordination and real-time availability management.

Dolphin Movie Downloader, Electron.js (December 2025)

- Developed a cross-platform desktop application for media streaming and downloads
- Implemented features for search, streaming, and local file storage
- Integrated auto-update functionality for seamless application maintenance

EDUCATION

Bachelor (Hons) in Information & Communication Technology,

2022 - 2026

Faculty of Technology University of Ruhuna

G.C.E. Advanced Level - Technology Stream,

2020

Central College, Wadduwa

CERTIFICATIONS

- Aviatrrix Certified Engineer, Multi-cloud Network Associate
 - Postman Student Expert, API Fundamentals
- Python for Beginners, University of Moratuwa
 - FreeCodeCamp, JavaScript Algorithms and Data Structures
- GitHub Foundations, Github
- AI/ML Engineer (Stage 1), SLIIT

ACHIEVEMENTS

- Hacktrail 1.0, ICT Students' Circle - University of Ruhuna

September 2024
- Hacktrail 2.0, ICT Students' Circle - University of Ruhuna

September 2025
- AdaptiQ Hackerthon, Information Institute of Technology

October 2025
- PITCH ARENA Hackerthon, Faculty of Engineering University of Ruhuna

December 2025

RESEARCH

- Cross-Platform Performance Modeling for Serverless Function Portability in Azure, AWS, GCP

January 2026
- Developing an ML-driven predictive tool (CLI/React Dashboard) based on 100k+ benchmarked invocations to forecast latency, costs, and cold starts across AWS, Azure, and GCP to mitigate vendor lock-in.

TECHNICAL SKILLS

Programming Languages & Frameworks Java, JavaScript (ES6+), Python, PHP, React.js, Next.js, Tailwind CSS,Node.js, Express.js, Spring Boot, Flutter AI / ML Machine Learning (basic models, CNN), Natural Language Processing (NLP), Image Processing, AI Tools & Productivity Languages & Soft Skills Sinhala, English, Team Leadership, Team Collaboration, Problem Solving	APIs & Architecture RESTful API Development, API Testing (Postman), Retrieval-Augmented Generation (RAG) Computer Science Fundamentals Software Engineering, Object-Oriented Programming (OOP), Design Patterns, Data Structures & Algorithms	DevOps & Cloud Docker, Kubernetes, CI/CD (GitHub Actions),Jenkins, Prompt Engineering, Multi-cloud Networking (AWS, Azure, GCP basics via Aviatrrix) Data Engineering & Analytics Data Warehousing (ETL pipelines), Data Modeling (Galaxy Schema), Power BI (basic visualization), Pandas (data processing)	Databases MySQL, MongoDB, PostgreSQL, MS Access Networking Fundamentals OSI Model, TCP/IP Model, IPv4 Addressing, Subnetting (basic), DNS, DHCP, HTTP/HTTPS, TCP vs UDP, Routing fundamentals, LAN vs WAN, Network devices (Router, Switch, Firewall), Basic network security concepts
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REFERENCES

- Mrs. R.D.N. Shakya, Lecturer (Probationary), Department of ICT, Faculty of Technology, University of Ruhuna
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- Mis. Chanduni Gamage, Lecturer (Probationary), Department of ICT, Faculty of Technology, University of Ruhuna
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